and the biodiversity within them. Terrestrial ecosystems, for example, can be subdivided into deserts, forests, grasslands, taigas, and tundras.



Tropical coral reefs are some of the most diverse ecosystems of all, full of fish, sea turtles, crustaceans, mollusks, and sponges, as well as corals.

Dynamic feedback

Tansley's key insight was that these discrete communities of living and nonliving components form dynamic systems. In a terrestrial ecosystem, for example, the organisms interact to recycle matter: plants absorb carbon dioxide (CO_2) from the atmosphere and nutrients from the soil to grow. These plants respire to release life-sustaining oxygen into the atmosphere, and provide food for animals. Animal excreta and dead matter also release